Investigating the Long-term Impacts of Hurricane Maria on Puerto Rican Households

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GSAW 2023

Organizing Decentralized Resiliency in Critical Interdependent Infrastructure Systems & Processes (ORDER-CRISP)

INTRODUCTION

Existing Vulnerabilities

Geographic	Infrastructural	Socio-economic
Isolated island	Poor quality	Economic instability
• Mountains in center, more people on coast • High climate vulnerability	• 'F' grade electricity infrastructure • 'D' grade water & transportation infrastructure	• >40% below poverty line • High debt • High social vulnerability

Hurricane Maria (Category 4, 2017)

• No evacuation option • Increased Physical exposure	• Extended utility disruption • Restoration challenge	• Employment loss • Permanent Migration

→ **Long-term impact**

OBJECTIVE

Assessing the long-term impact of Hurricane Maria on Puerto Rican households (HH) to analyze the role of storm-induced damages and existing socio-economic and environmental factors.

METHODS

- Storm Characteristics
- Social vulnerability
- Geography
- Puerto Rico (2021)
- Total response: 1804

Geospatial Data → Spatial Analysis (• Exposure at spatial levels • Distance from spatial levels)

Household Survey → Statistical Analysis → Descriptive Statistics; Ordered Logit Regression

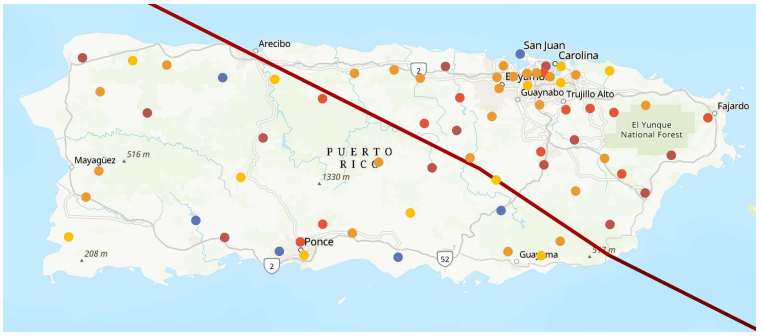
Long-term Impact

- Damage & Exposure
- Protective Actions
- Displacement
- Utility Disruption
- Household Characteristics
- Disaster Relief

RESULTS

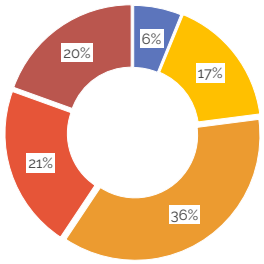
Household Location

- 502 Households (72 zip codes)
- Low-income households: 100%
- 88% located <5 miles from the shore
- Maximum gust: 133 mph
- 34% respondent houses flooded
- 45% received aids/ assistance



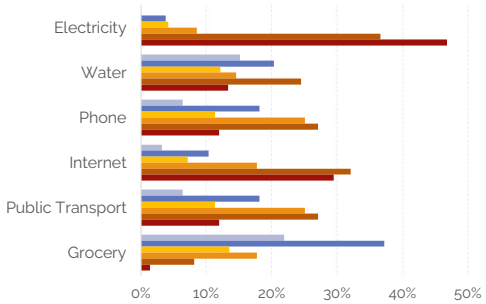
No Impact · Low Impact · Some Impact · High Impact · Extreme Impact

Household Impact



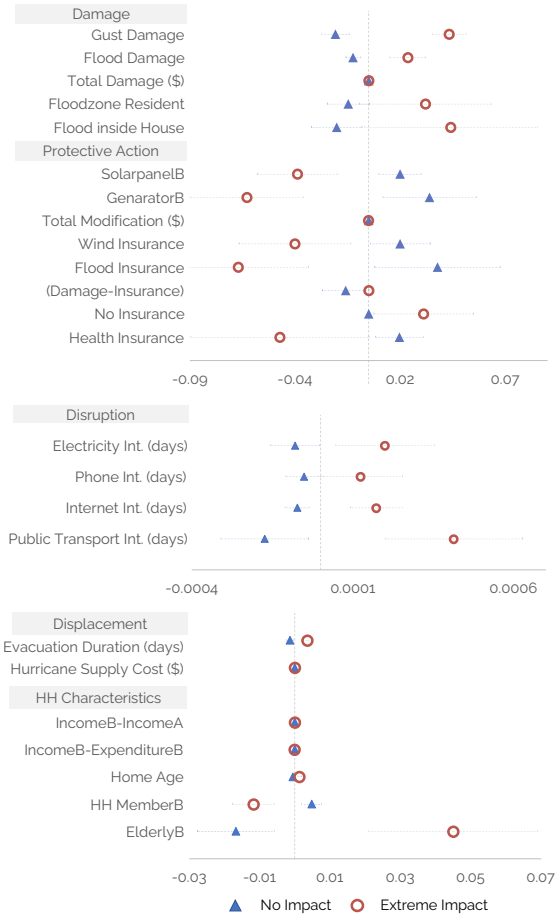
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Utility-Service Interruption



No interruption · 1-7 days · 8-14 days · 15-30 days · 31-90 days · > 3 months

Marginal Effect



IMPLICATIONS

These research findings may be useful for improving disaster management and reducing long-term impacts in vulnerable regions.

References:

DISCUSSION

- Encountering damage from hurricane gusts caused an intensive impact compared to flood damage.
- Essential infrastructure like water supply disruption and grocery closure period had no significant relation to HH impact for extended period.
- Fast restoration of communication infrastructure, especially internet can improve household well-being.
- Despite fuel shortage, HH owning generator faced less impact than HH owning solar panel.
- Disaster relief was insufficient to negate hurricane impact-raising HH dissatisfaction.
- Over 50% HH whose income reduced following Hurricane Maria had not recovered their pre-hurricane income level.
- Larger HH overall faced less extended impact despite elderly members.